08-19 第2篇

阅读理解（说明文） How Does a Spider Build Such a Perfect Web?

① On a cool autumn morning, have you ever stopped in front of a garden gate to look at a **spider's** web covered with tiny drops of mist? Each thread shines in the early light; the whole pattern looks almost like a silver wheel. Then you notice one small brown spider sitting quietly in the middle, and you may ask: **how did this tiny creature, with no ruler and no plan, build such a neat and strong home?** Scientists have studied spider webs for more than a hundred years, and the story behind each web is surprisingly clever.

② The first thing to know is **where a spider's silk comes from**. A spider has several tiny body openings called **spinnerets** at the end of its body. Inside the spider, there are small **glands** that make a sort of **liquid protein**. When the spider needs silk, it pushes this liquid out of its spinnerets. As soon as the liquid meets the air, it **hardens** into a very thin thread. A spider can actually produce **several different kinds of silk** — some sticky, some strong, some stretchy — and it uses each kind in different places of the web.

③ The second thing to know is that the spider does **not** weave a web in random circles. A typical garden spider follows a clear order. Step 1 is called "**the first bridge**". The spider stands on a high point and lets a thin silk thread fly out into the wind. If the thread touches another branch and sticks, the spider has built its first bridge. If it fails, it eats the thread back and tries again.

④ Step 2 is "**the Y-shape frame**". Once the first bridge is ready, the spider walks slowly along it, making it thicker and stronger. Then it drops down from the middle of this thread with a second silk line, and attaches it to a lower branch. Now the web looks like a rough letter "Y". This Y-shape is the skeleton of the whole web.

⑤ Step 3 is "**the outer frame and the straight threads**". The spider then walks around the Y, adding an outer frame, and builds many straight silk threads, all going out from the centre like the lines of a bicycle wheel. These straight threads are **not sticky**. The spider uses them as its own walking paths.

⑥ Step 4 is the most beautiful part — "**the sticky spiral**". Starting from near the centre, the spider walks round and round on its non-sticky straight threads, laying a thin sticky thread in a big slow spiral from the outside to the middle. The spider actually puts small drops of glue on this spiral as it goes. These sticky drops are what catch the flying insects.

⑦ Finally, the spider rests, usually in the very middle or just under a leaf, waiting for the web to move. When a small insect flies into the sticky spiral, the whole web shakes. The

spider feels this shake through its legs and quickly runs out along the safe straight threads to catch its meal. It remembers, without thinking, which threads are sticky and which are safe. That is why you almost never see a spider stuck on its own web.

⑧ A spider usually builds a whole new web in just **thirty to sixty minutes**, most often at night. Old webs get dusty or broken, so many spiders even **eat their old web first**, to re-use the protein. That way, nothing is wasted. Next time you see a spider web in the garden — tiny, shining, with silver threads in perfect order — remember that you are looking at a small engineer who has quietly built a home, a kitchen and a fishing net, all in one early morning.

注释： spinneret 喷丝口；gland 腺；protein 蛋白质；spiral 螺旋

根据文章内容，选择最佳答案。

1. Where does a spider's silk come from?

A. From its eyes.
B. From special glands inside the body and tiny spinnerets at the end of the body.
C. From a small hole on its head.
D. From its ears, hidden under the legs.

2. What is the first step when a garden spider starts to build a web?

A. Draw a large Y-shape on a paper.
B. Let a thin silk thread fly out in the wind to build a "first bridge" to another branch.
C. Build the outer frame first.
D. Lay the sticky spiral in the middle first.

3. Why does the spider rarely get stuck in its own web?

A. Because its legs are made of glass.
B. Because it remembers which straight threads are safe (not sticky) and walks only along those.
C. Because its body is covered in oil.
D. Because the whole web is actually not sticky.

4. What does the passage say about old, broken webs?

A. The spider throws them into the river.
B. Many spiders eat their old webs so that the silk's protein can be re-used, and nothing is wasted.
C. The wind always blows them all away.
D. Humans collect them for making silk cloth.

5. What is the writer's main attitude toward spiders in the passage?

A. Fear and hate — spiders are dangerous and should all be killed.
B. Admiration — spiders are clever little engineers who build a beautiful and useful home.
C. Indifference — the writer does not really care about spiders.
D. Pity — the writer feels very sorry for all spiders.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "spinnerets at the end of its body ... small glands that make a sort of liquid protein"
【解析】原文明确指出丝来自身体内部的腺体，经喷丝口吐出。A、C、D 均无依据。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "the first bridge ... lets a thin silk thread fly out into the wind"
【解析】Step 1 是借风搭起第一根桥线。A、C、D 均与步骤顺序不符。故选 B。

3.
【答案】B
【题型】推理判断题
【定位】第⑤段 "These straight threads are not sticky. The spider uses them as its own walking paths" + 第⑦段 "which threads are sticky and which are safe"
【解析】直线丝不粘，蜘蛛只走这些安全线，因此不被粘住。A、C、D 均无文中依据。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑧段 "many spiders even eat their old web first, to re-use the protein. That way, nothing is wasted"
【解析】原文直接说明蜘蛛会吃旧网回收蛋白质。A、C、D 均与原文不符。故选 B。

5.
【答案】B
【题型】写作态度题
【定位】第⑧段末 "a small engineer who has quietly built a home, a kitchen and a fishing net, all in one early morning"
【解析】作者把蜘蛛称为小工程师，语气满是钦佩。A、C、D 均与作者语气不符。故选 B。